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LONG TERM FORECASTING FOR STRONG PERIODIC SEQUENCE

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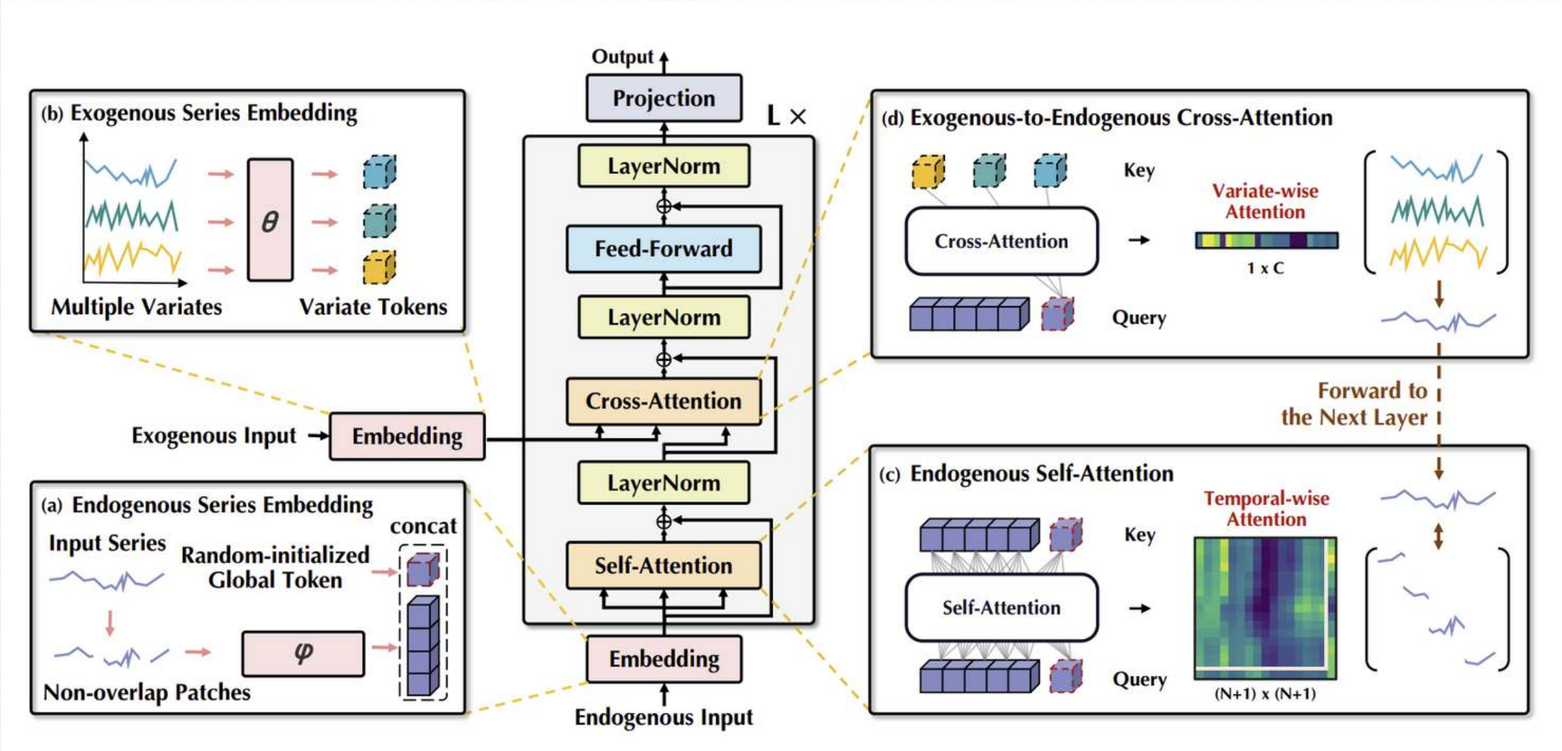
Introduction

- Most existing models follow a "one-size-fits-all" approach for general time-series, often failing to capture intricate temporal patterns in complex datasets

- We propose a specialized forecasting model **TaXer/WaXer** which explicitly leverages these periodicities

- We construct a real-world weather dataset **MeteoPolitan** from major global cities, providing a robust benchmark for evaluating periodicity-driven time-series models.

Preliminaries - TimeXer

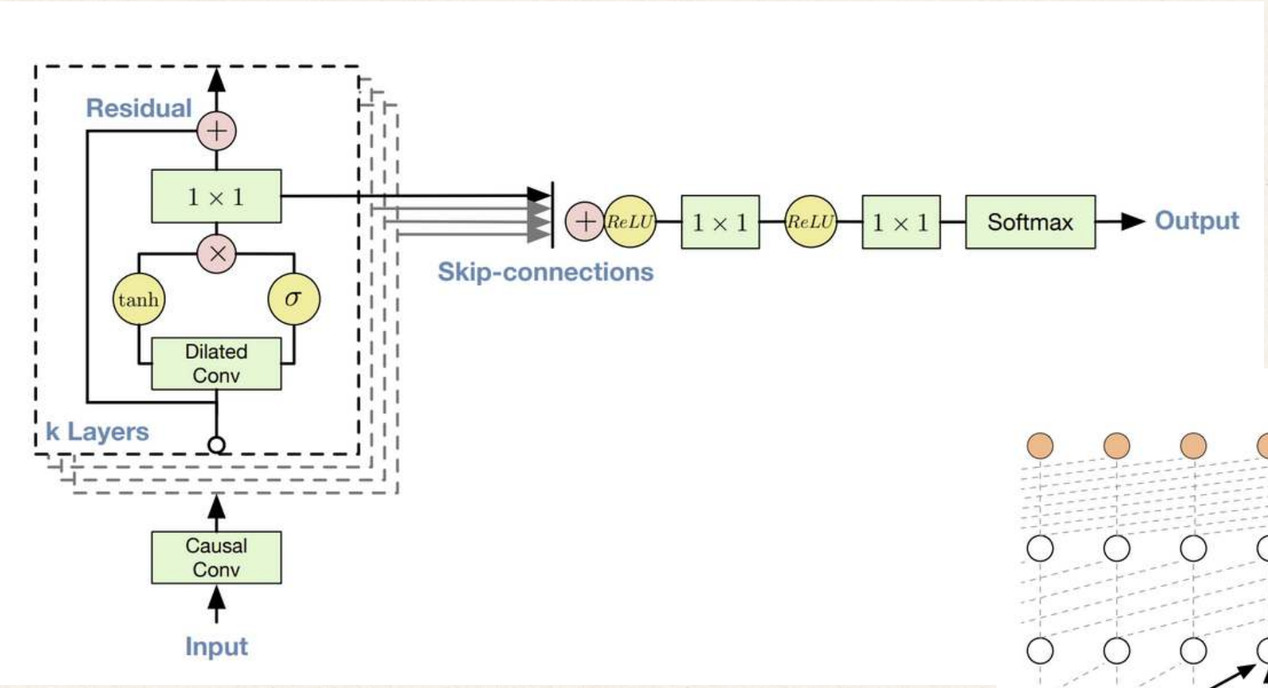


Architecture of TimeXer

- A Transformer-based **SOTA model** optimized for time-series prediction
- Embedding Strategy
  - Endogenous input
    - Segmented into patch-level tokens
  - Exogenous input
    - Embedded as series-wise variate tokens
- Self-attention among endogenous tokens to capture temporal dependencies
- Cross attention with exogenous tokens to selectively integrate external information

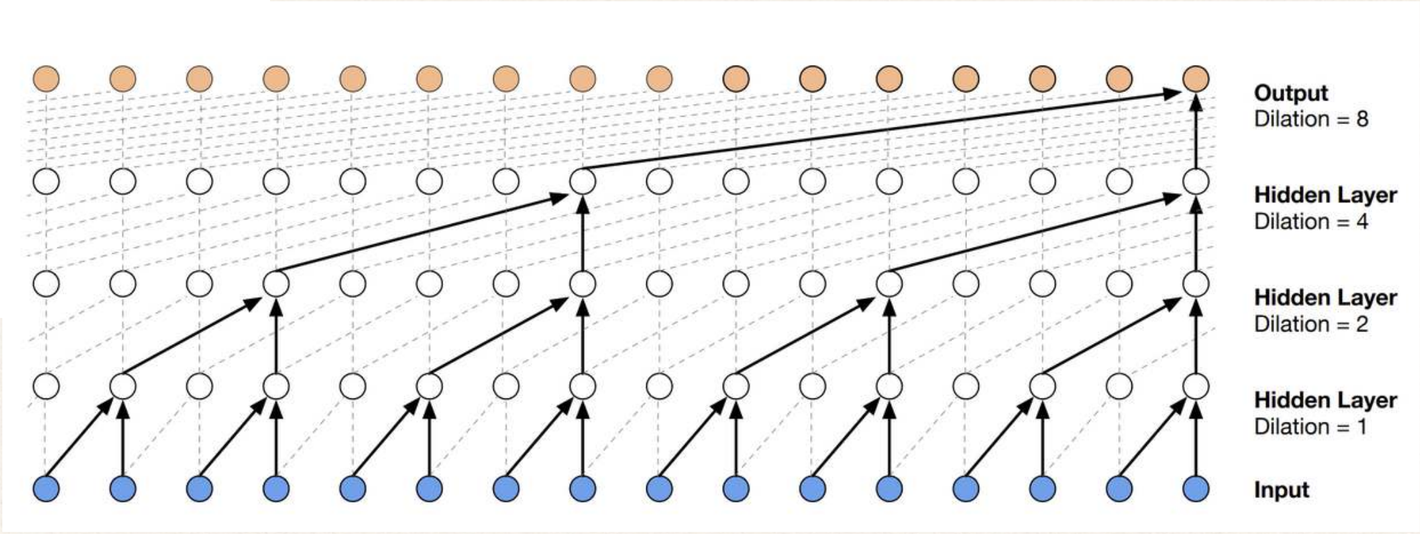
Y. Wang et al., "TimeXer: Empowering Transformers for Time Series Forecasting with Exogenous Variables," NeurIPS, 2024.

Preliminaries - WaveNet



Architecture of WaveNet

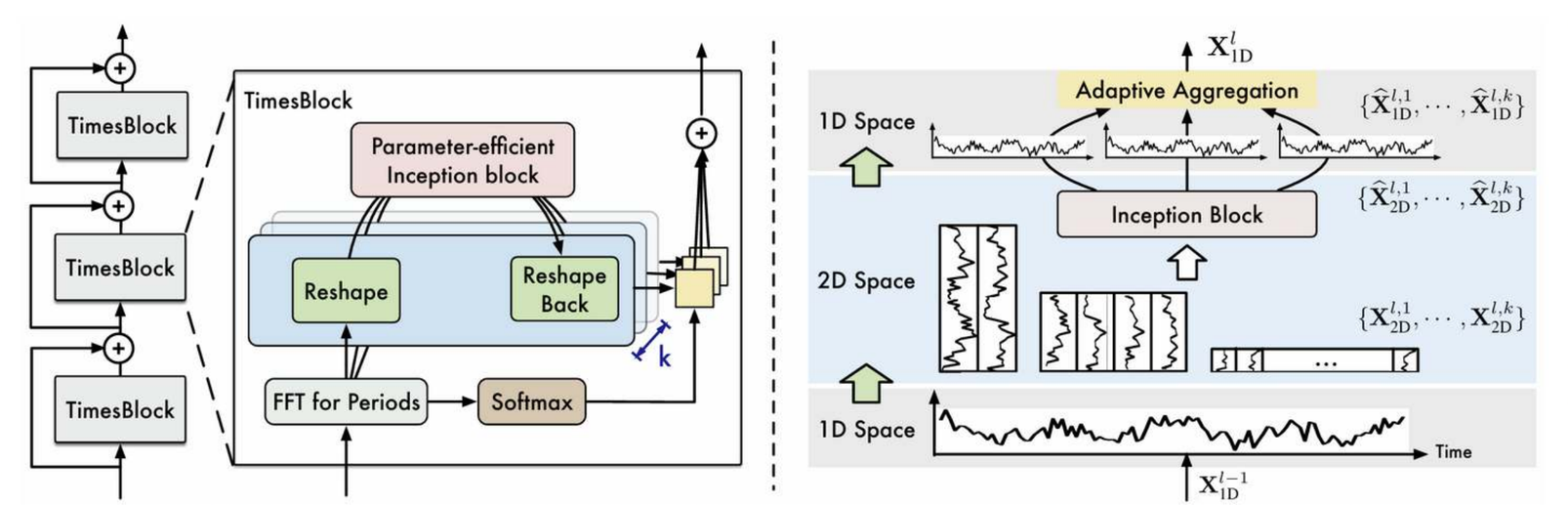
- Autoregressive model that generates raw audio waveforms
- Use dilated convolution which exponentially increases the receptive field to capture long-range audio context
- Dilation factor doubles at every layer up to a limit and then repeats from 1 (e.g., 1, 2, 4, ..., 512, 1, 2, ...)



Visualization of dilated causal convolutional layers

A. van den Oord et al., "WaveNet: A Generative Model for Raw Audio," arXiv:1609.03499, 2016.

Preliminaries - TimesNet

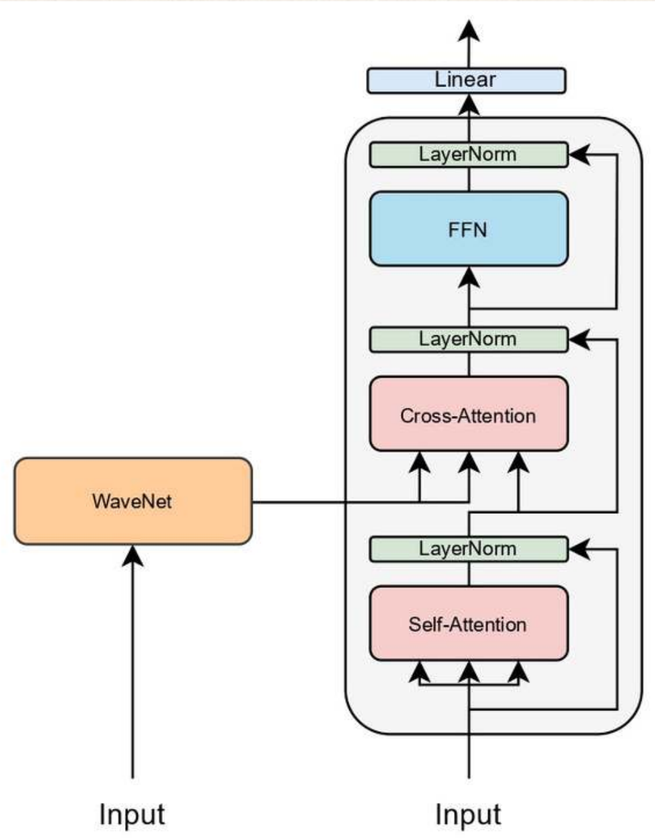


Architecture of TimesNet

- Identify top-k frequencies to determine the most dominant period
- Segment the 1D series based on detected period and stacks them horizontally
  - Intra-period(vertical)
    - Capture short-term temporal variations within a single period
  - Inter-period(Horizontal)
    - Capture long-term trend across different periods
- Use Inception-based 2D Convolution block

H. Wu et al., "TimesNet: Temporal 2D-Variation Modeling for General Time Series Analysis," ICLR, 2023.

Method - WaXer



Architecture of WaXer

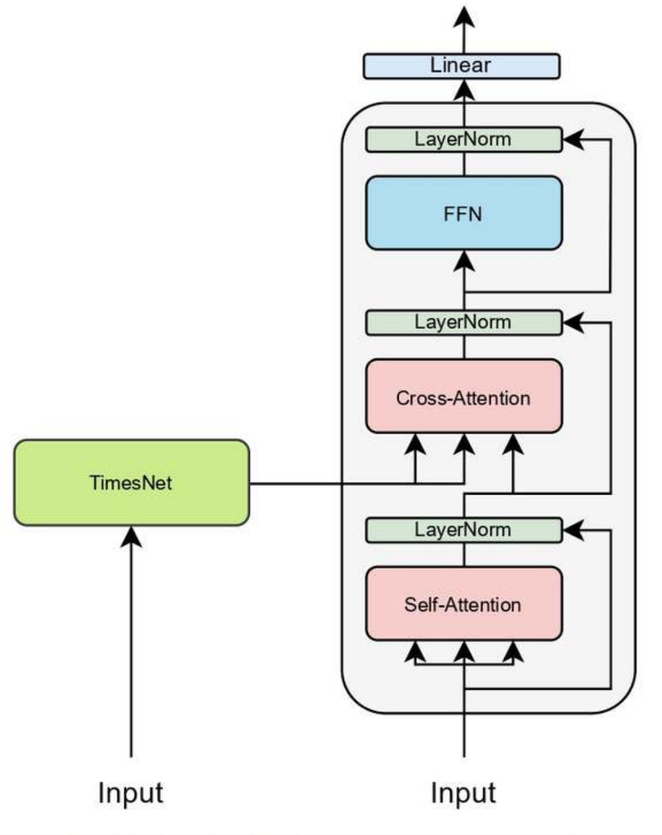
WaveNet as a Feature Extractor

- Dilated Causal Convolution
  - Captures long-range temporal dependencies efficiently
  - Maintains causality (no future information leakage)
- Residual & Skip Connections
  - Enables training of deep networks
  - Aggregates multi-scale features

Integration Mechanism

1. WaveNet extracts temporal features from raw input
2. Linear projection aligns WaveNet features to TimeXer dimension
3. Cross-attention in encoder layers:
   - Query: TimeXer's global token
   - Key/Value: WaveNet features
4. Flatten head produces final prediction

Method - TaXer



Architecture of TaXer

TimesNet as a Feature Extractor

- FFT-based Period Detection
  - Applies FFT and identifies top-K dominant frequencies
  - Determines period lengths automatically
- Extracting inter-period features
  - Slice the sequence with highest FT signal
  - 2D analysis can extract inter-period relation

Integration Mechanism

1. TimesNet extracts periodic features via 2D convolution
2. Linear projection aligns TimesNet features to TimeXer dimension
3. Cross-attention in encoder layers:
   - Query: TimeXer's global token
   - Key/Value: TimesNet features
4. Flatten head produces final prediction

Experiments: Setup

Hardware Spec.

CPU	2x AMD EPYC 7763 64-Core
GPU	8x NVIDIA RTX A5000
CUDA	12.2
Driver	535.171.04
OS	Ubuntu 20.04

Key packages

Pytorch	2.8.0
NumPy	1.26.4
Pandas	2.3.3
scikit-learn	1.6.1

Baselines

iTransformer (ICLR 2024)
TimeXer (NeurIPS 2024)

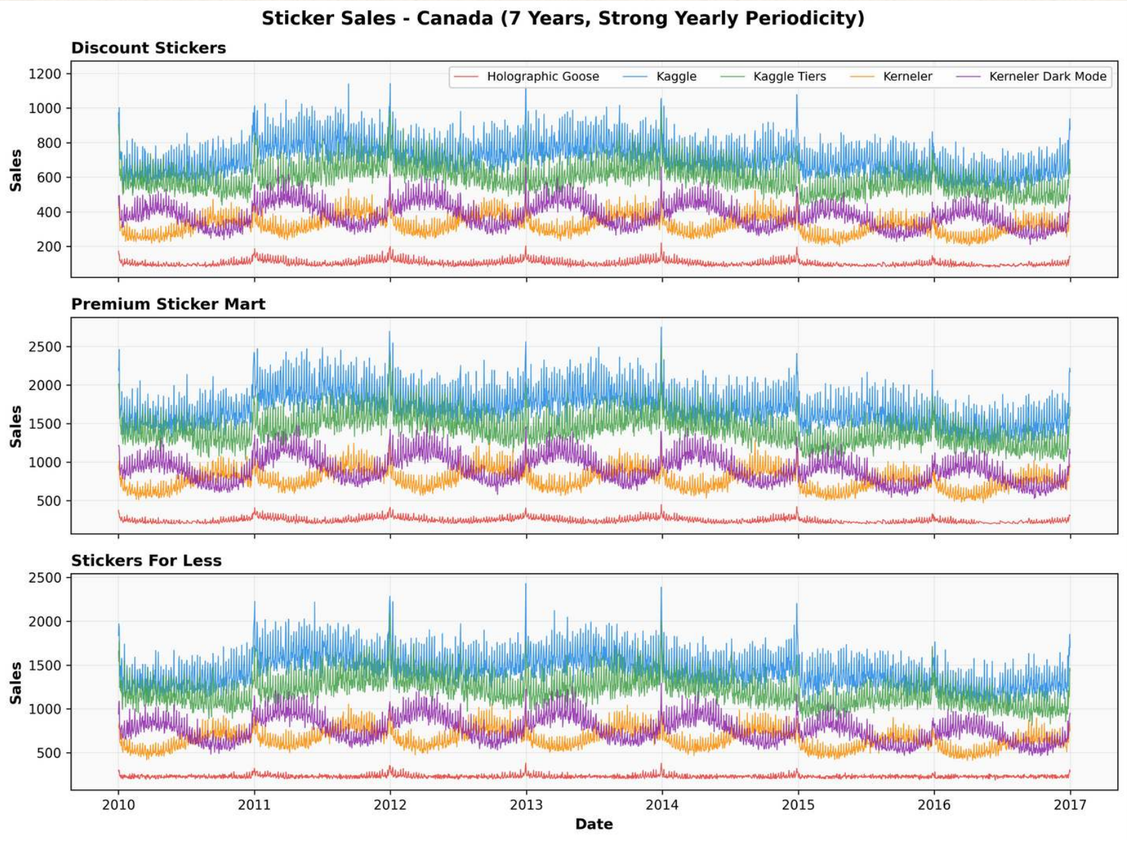
Metric

MSE	Mean Squared Error (prior metric for LTSF)
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
RMSE	Root Mean Squared Error
CORR	Pearson Correlation

Key hyperparameters

seq_len	512	Input lookback window
pred_len	16	Single forward pass prediction
d_model	64	Embedding dimension
d_ff	128	FFN dimension
n_heads	4	# of attention heads
e_layers	2	# of transformer layers
batch_size	32	batch size
optimize	-	AdamW(lr=1e-4, beta=[0.9,0.99])
epochs	128	# of training epochs

Experiment: Dataset

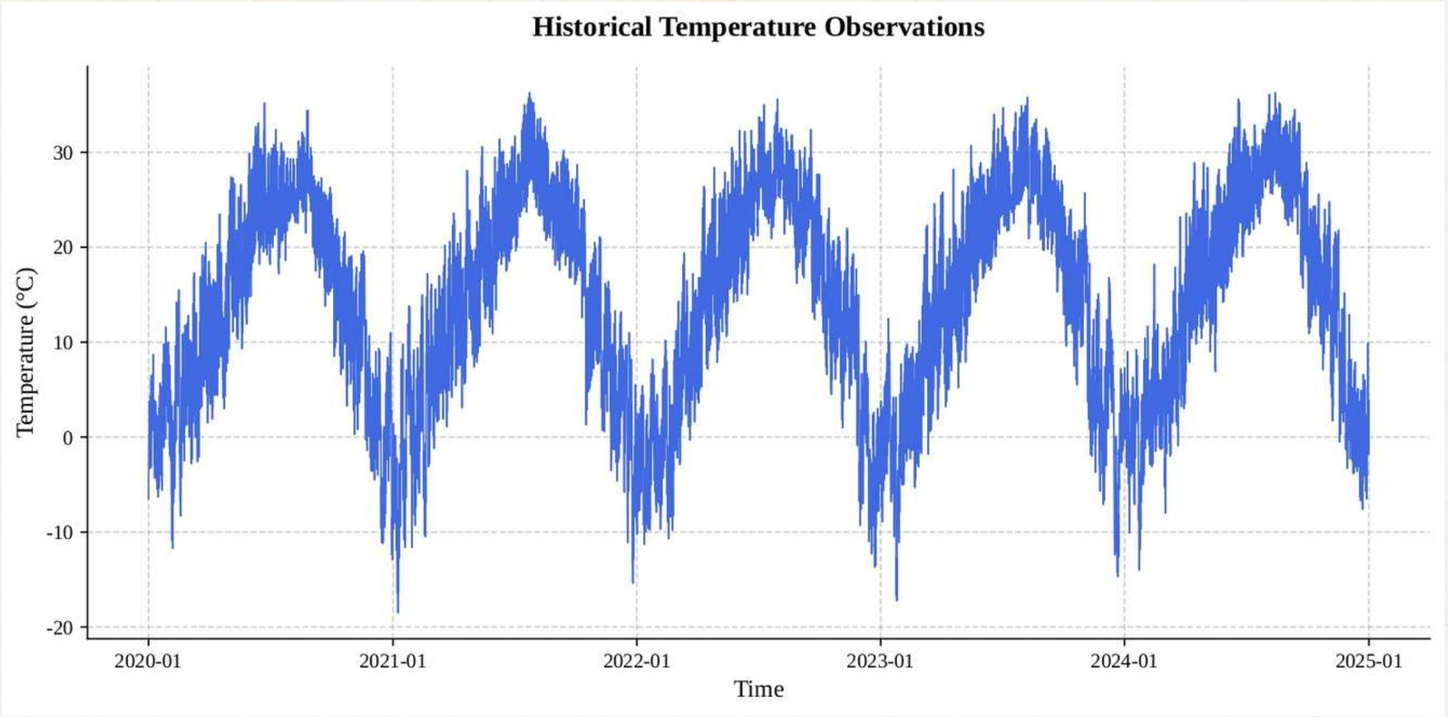


Time-series Variation of Sticker Sale Volume

- Dataset Overview
  - Sticker Sales data with 7-year.
  - Strong periodicity observed.
- 6 Countries
  - Canada
  - Finland
  - Italy
  - Kenya
  - Norway
  - Singapore
- 3 stores per country
- 6 product categories per store
- Total Time Series: 6 x 3 x 5 = 90

- data format: csv
- data type: float32
- datapoint unit: day

Experiment: Dataset



Time-series Variation of Hourly Temperature in Seoul

- Construct Korea & Global weather datasets by selecting five key variables
  - Features are identified through ACF analysis as having the highest periodicity
  - Temperature, Wind Speed, Wind Direction, Sea Level Pressure, and Humidity/Dew Point
- All data from 2020 to 2024 (5 years) were collected from official national meteorological agencies and strictly aligned into a unified format to ensure consistency across different regions.
- Ensure dataset reliability by removing low-quality records and apply linear interpolation for missing values

Experiments: results

Global Weather

Model	MSE	MAE	RMSE	MAPE	CORR
iTransformer	0.6491	0.7387	0.8595	2.8568	0.4564
TimeXer	0.6255	0.6614	0.8132	2.0352	0.4944
WaXer	0.6253	0.6624	0.8139	2.0838	0.4882
TaXer	0.6135	0.6474	0.8046	2.3100	0.5202

Experiments: results

Korean Weather

Model	MSE	MAE	RMSE	MAPE	CORR
iTransformer	0.5796	0.6093	0.7806	2.9365	0.5143
TimeXer	0.5505	0.5290	0.7273	2.3945	0.5544
WaXer	0.5515	0.5312	0.7288	2.4390	0.5504
TaXer	0.5413	0.5221	0.7226	2.3204	0.5587

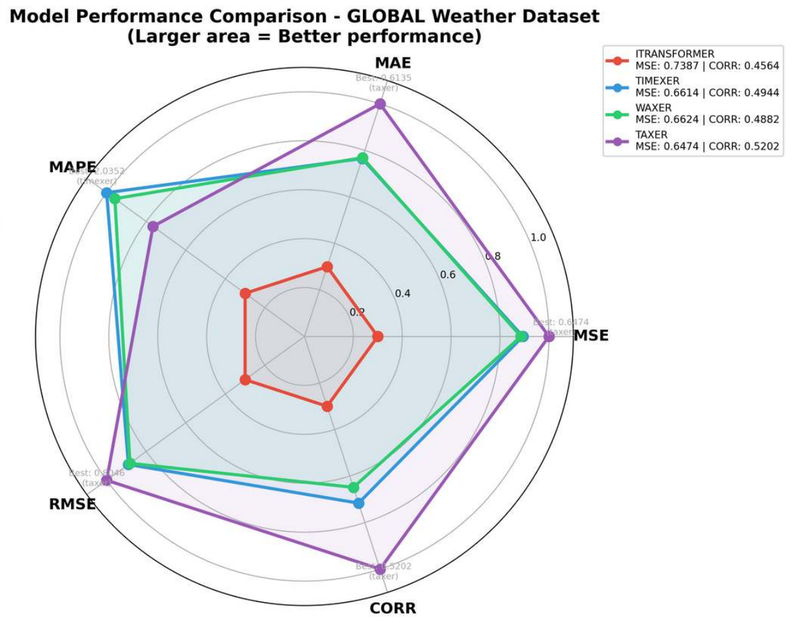
Experiments: results

Stickers

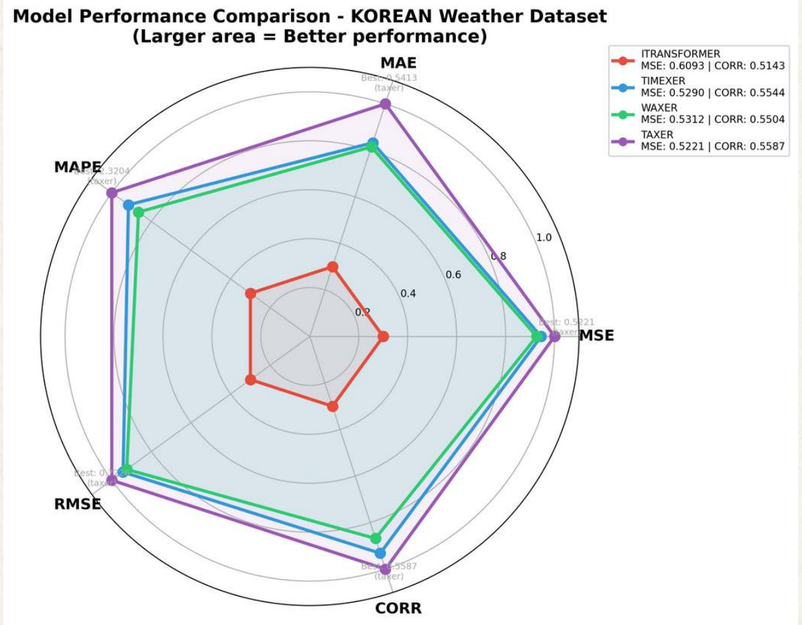
Model	MSE	MAE	RMSE	MAPE	CORR
iTransformer	0.2495	0.3715	0.4995	1.7889	0.7368
TimeXer	0.1647	0.3077	0.4058	1.1816	0.8152
WaXer	0.1609	0.3052	0.4012	1.1923	0.8239
TaXer	0.1569	0.2971	0.3961	1.1868	0.8196

Experiments: results

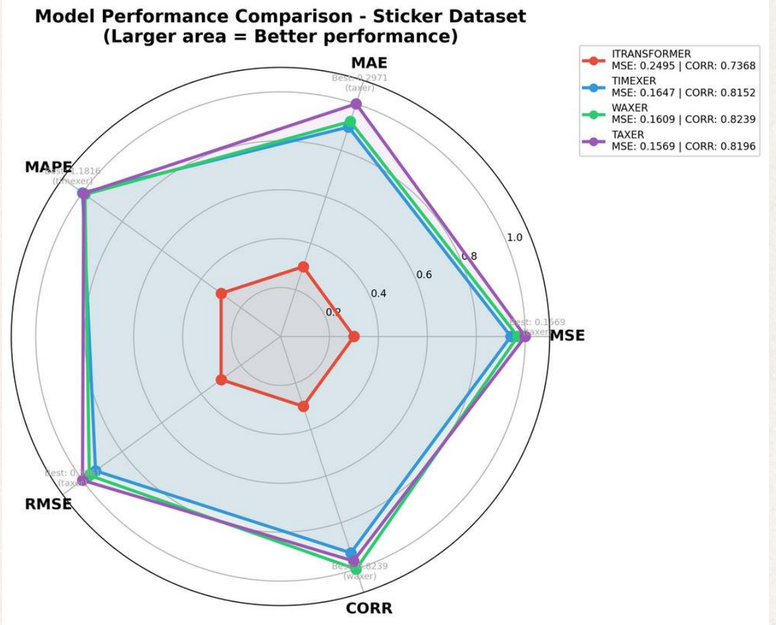
Global Weather



Korean Weather

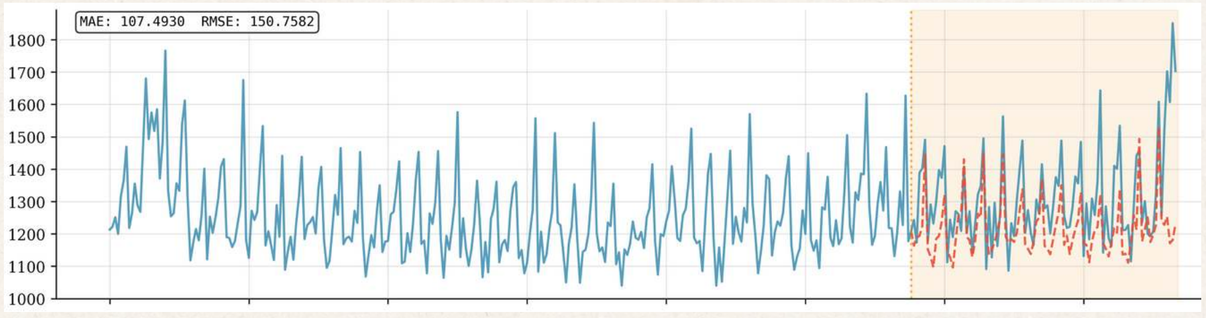


Stickers

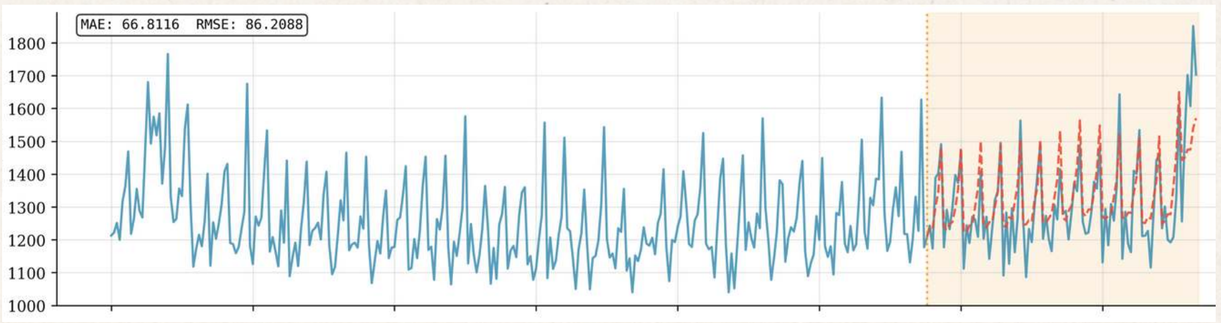


Experiments: results

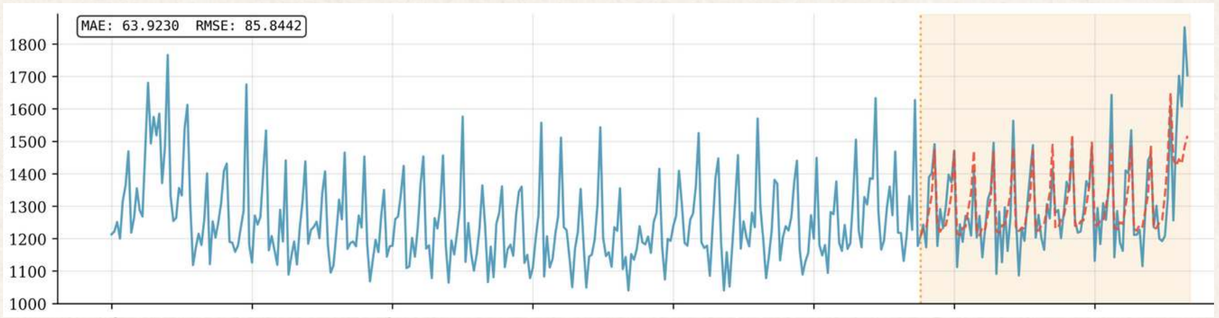
iTransformer



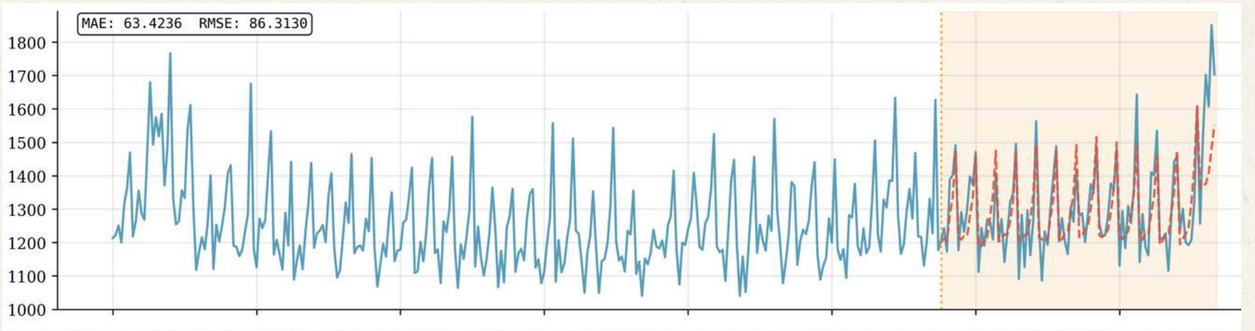
TimeXer



WaXer

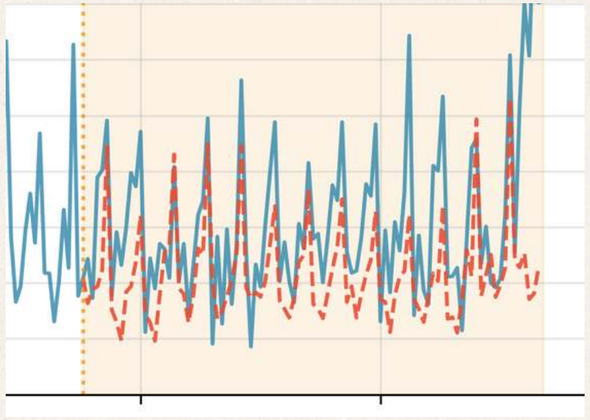


TaXer

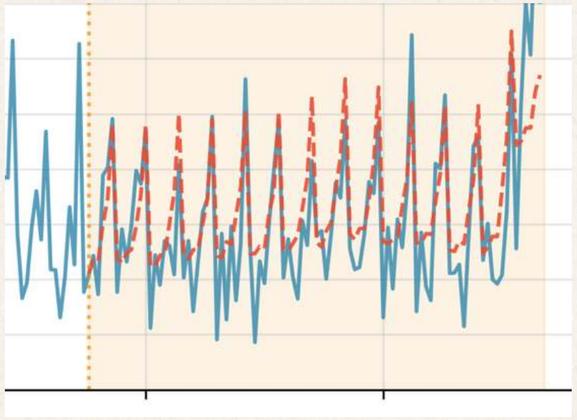


Experiments: results

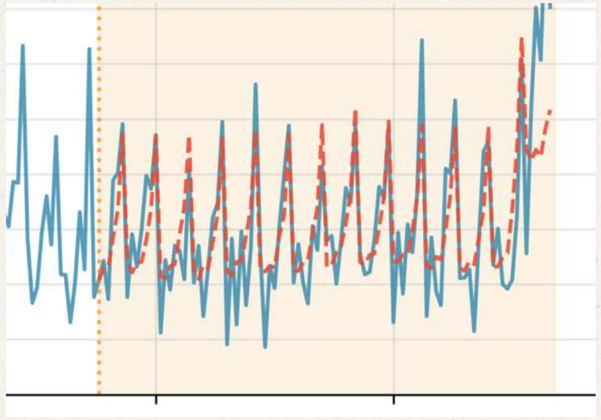
iTransformer



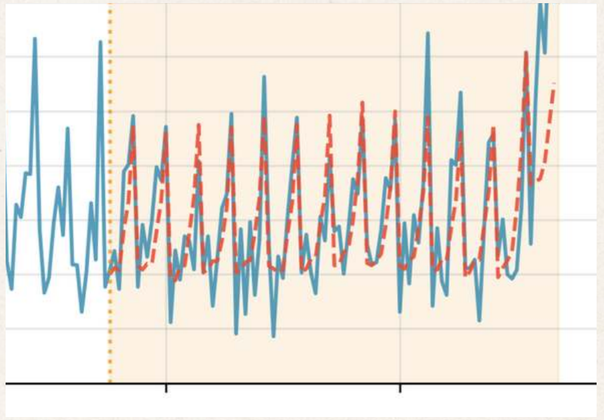
TimeXer

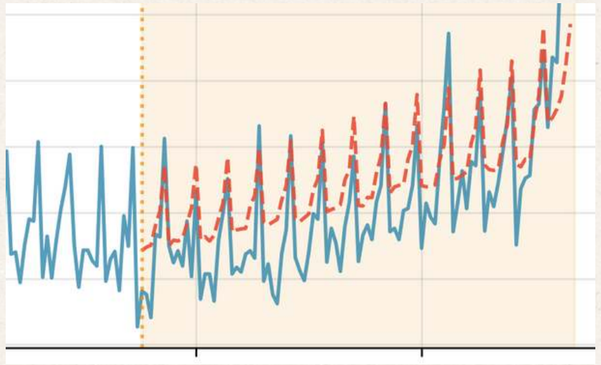


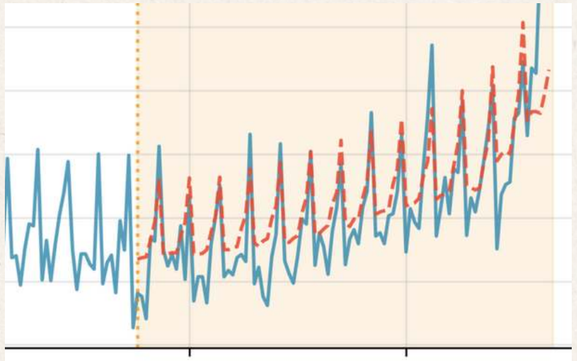
WaXer

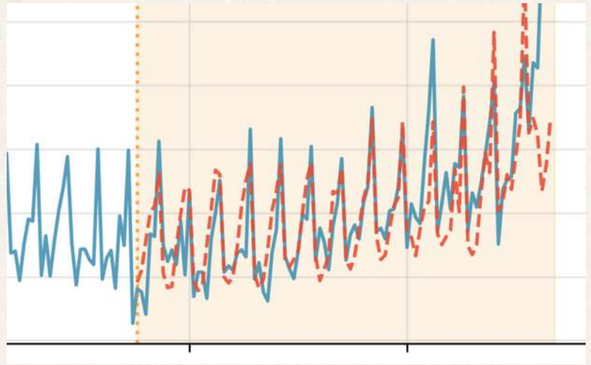


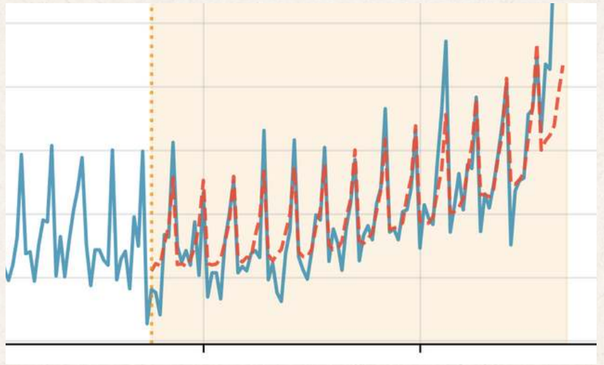
TaXer











Conclusion

Model Improvements

- **TaXer achieves the best performance** across all datasets
  - Sticker: 37% lower MSE than iTransformer, 4.7% lower than TimeXer
  - Global Weather: 12% lower MSE than iTransformer, 2.1% lower than TimeXer
  - Korean Weather: 14% lower MSE than iTransformer, 1.3% lower than TimeXer
- **Hybrid models consistently outperform baselines**
  - TaXer and WaXer beat both TimeXer and iTransformer on every dataset
  - Combining transformer attention with specialized feature extraction is effective

Architecture Insights

- **Feature extraction before attention is beneficial**
  - Both TaXer and WaXer improve upon TimeXer baseline
  - Pre-processing with temporal feature extractors enhances transformer performance
- **TaXer / WaXer trade-off**
  - TaXer (TimesNet features): Better MSE/RMSE scores, longer inference due to FFT
  - WaXer (WaveNet features): Second place performance, over 200% faster than TaXer